ReMis [A fun title could go here]

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Introduction

Structural equation models (SEMs) are widely used in applied research to test hypotheses about latent constructs and their relationships. In practice, replication is typically assessed by fitting the same model to a new dataset and evaluating whether global fit indices remain acceptable or whether selected parameters retain statistical significance. Empirically, these approaches often yield ambiguous or unstable outcomes. Models may exhibit acceptable global fit across datasets while differing substantially in parameter estimates, reflecting the non-uniqueness of the model-implied covariance structure. Conversely, parameters that were statistically significant in an original study may fail to reach significance in a replication due to differences in sample size, measurement quality, or sampling variability. In addition, estimation itself may be unstable. A model that converges in one dataset may fail to converge in another sample drawn from the same population, or even in resampled versions of the original data. Such behavior indicates sensitivity of the model to sampling fluctuations or weak identification, and implies that replication may fail at the level of model estimation rather than parameter comparison. These issues highlight, that replication in SEM is underdetermined and unsolved. To solve that problem, we argue that replication of SEMs can be formalized as a problem of parameter stability analogous to invariance testing.

Replication

Replication is the ability to obtain consistent findings when a study is repeated with new independent data being collected under the same or sufficiently similar methodological and theoretical conditions as the original study. In the context of empirical research, a finding can be understood at different analytical levels: as an empirical regularity, as an estimand-based claim, or as a theoretical claim about an underlying mechanism that generates observed patterns. The type or extent of replication relevant to a finding depends on which of these levels the finding is defined at.

Radder (1992) characterizes replication as comprising three distinct forms: (a) material replication; (b) reproducibility of the theoretical interpretation; and (c) replication of the experimental result through an alternative material procedure. Schmidt (2009) builds on this definition and establishes the concepts of direct replication as a restrictive and narrow form of replication and conceptual replication as a wider notion of replication.

Let $d \in \{1, 2\}$ index datasets, with $d = 1$ denoting the original study and $d = 2$ the replication study. In direct replication, the same estimand θ^* is targeted under equivalent conditions:

$$\theta^{*(1)} \approx \theta^{*(2)}.$$

Under finite sampling, exact equality,

$$\theta^{*(1)} = \theta^{*(2)},$$

is generally implausible.

Thus, direct replication refers to a repetition of the original experimental or observational procedure aimed at re-estimating the same estimand under equivalent conditions. This in turn relates to non-experimental methods as when a new study estimates the same estimand under conditions that are equal to the original, operationalizing the variables in the same way, with differences limited to incidental sources of variation such as sample, time, or setting. Under this definition, deviations from exact numerical equality of coefficients are to be minimal, however exact numerical equality is not neccessary.

By contrast, conceptual replication concerns the stability of a theoretical claim under changes in operationalization. This relates to Radder (1992), his b and c compartments of a replication, so that the same theoretical claim or construct is supported using a different estimand that is theoretically analogous to the original. This involves changes in operationalization, such as different independent or dependent measures, while retaining the underlying conceptual target. Let $\theta^{(1)} \in \Theta_1$ and $\theta^{(2)} \in \Theta_2$ denote estimands from two differently operationalized studies. Conceptual replication can then be expressed through a mapping ϕ such that

$$C(\theta^{(1)}) = C(\phi(\theta^{(2)})),$$

where $C(\cdot)$ denotes the theoretical claim supported by the estimand. In this case, replication is evaluated at the level of theoretical equivalence rather than parameter identity.

Replicating Structural Equation Models

According to Kaplan (2009), SEM comprises a class of methodologies designed to express a hypothesized model through summary statistics derived from empirical observations, parameterized by a reduced set of structural parameters. As SEMs are also considered to be a way to formulate social science theories(De Leeuw and Berk 2009), for a SEM to be deemed replicated, the encompassed theory needs to hold in the data, either globally or regarding certain parameters.

For models involving continuous latent variables, the associated hypotheses typically pertain to the means, variances, and covariances of the observed variables. Hypotheses may also concern effect sizes, ordering of parameter magnitudes, or specific parameter values. Because SEMs often contain many estimated parameters, replication should generally focus on those outcomes that are substantively relevant to the hypothesis under study.

Global model fit versus parameter-level replication

Global model fit concerns the adequacy of the model-implied covariance structure. Let $\Sigma(\theta)$ denote the covariance matrix implied by the model parameters θ, and let $\mathbf{S}^{(d)}$ denote the observed covariance matrix in dataset d. Global fit evaluates whether

$$\Sigma(\theta) \approx \mathbf{S}^{(d)}.$$

Equivalently, one may express this through a fitting function F such that replication at the level of global fit would require

$$F\big(S^{(d)}, \Sigma(\hat{\theta}^{(d)})\big) \leq c \quad \text{for } d = 1, 2,$$

where c denotes a context-dependent criterion for acceptable model fit.

However, good global fit in both datasets does not imply replication at the level of parameters. In general,

$$\Sigma(\theta^{(1)}) \approx \Sigma(\theta^{(2)})$$

does not imply

$$\theta^{(1)} \approx \theta^{(2)}.$$

Multiple parameter configurations may produce similar model-implied covariance structures. Global fit therefore concerns the adequacy of the model-implied distribution, whereas replication at the level of parameters concerns the stability of theoretically relevant estimands.

Accordingly, restricting attention to estimands that correspond directly to model parameters,let $\theta^* := (\theta_{j_1}, \dots, \theta_{j_r})$ denote the subset of model parameters corresponding to the substantive claim of interest.

Replication at the level of parameters is defined as

$$\theta^{*(1)} \approx \theta^{*(2)}.$$

This formulation captures replication as stability of theory-relevant parameters, rather than mere admissibility of the overall model.

If i aim at replication it is therefore reasonable to treat both datasets as independent realizations of the same underlying model. Thus, it is a problem of symmetric parameter comparison across datasets. In a multiple-group SEM framework, parameters are estimated freely in both datasets and then compared through equality constraints imposed across groups. This reflects the null hypothesis

$$H_0 : \theta^{*(1)} = \theta^{*(2)},$$

which is formulated at the level of population parameters rather than sample-specific estimates. Empirically, this hypothesis is assessed using the corresponding parameter estimates. Because exact equality is often too strict a criterion, approximate formulations are more plausible, for example

$$\left|\widehat{\theta}_j^{*(1)} - \widehat{\theta}_j^{*(2)}\right| \leq \epsilon_j \quad \forall j,$$

where ϵ_j denotes a substantively justified tolerance for parameter j.

Under this perspective, replication of SEMs can be treated as analogous to invariance testing across groups, with datasets taking the role of groups. In the context of multiple-group SEM, datasets are treated as distinct groups. Accordingly, we define $g \equiv d$, where g indexes groups and d indexes datasets. This analogy can be formalized using the framework of measurement and structural invariance.

Invariance Testing

Measurement invariance (MI) refers to the statistical property that a measurement instrument assesses the same latent construct equivalently across different groups. In other words, the measurement model operates in the same manner across groups. When MI is established, meaningful comparisons of the underlying construct between groups are justified.

Let $\mathbf{y}_i^{(g)}$ denote the observed variables for individual i in group g, and let $\eta_i^{(g)}$ denote the corresponding latent variables. The measurement model can be written as

$$\mathbf{y}_i^{(g)} = \nu^{(g)} + \Lambda^{(g)}\eta_i^{(g)} + \varepsilon_i^{(g)},$$

with

$$\varepsilon_i^{(g)} \sim (0, \Theta^{(g)}).$$

Where:

- $\mathbf{y}_i^{(g)}$: observed responses of individual i in group g

- $\nu^{(g)}$: item intercepts
- $\eta_i^{(g)}$: latent variables for individual i
- $\Lambda^{(g)}$: factor loadings linking latent variables to observed indicators
- $\varepsilon_i^{(g)}$: residual errors
- $\Theta^{(g)}$: residual covariance matrix

From a statistical modelling perspective, MI refers to invariance of the measurement parameters across groups:

$$\theta_M^{(g)} = \theta_M \quad \forall g,$$

where

$$\theta_M = \{\Lambda, \nu, \Theta\}.$$

Structural invariance (SI), by contrast, refers to equality of structural coefficients across groups, that is, to the requirement that the same association between latent factors exists across groups. The structural model may be written as

$$\eta_i^{(g)} = \alpha^{(g)} + \mathbf{B}^{(g)} \eta_i^{(g)} + \zeta_i^{(g)},$$

with

$$\zeta_i^{(g)} \sim (0, \Psi^{(g)}).$$

where

- $\alpha^{(g)}$ contains latent intercepts
- $\mathbf{B}^{(g)}$ contains the structural coefficients
- $\zeta_i^{(g)}$ is a vector of structural disturbance terms
- $\Psi^{(g)}$ is their covariance matrix

SI is then defined as

$$\theta_S^{(g)} = \theta_S \quad \forall g,$$

with

$$\theta_S = \{\mathbf{B}, \alpha, \Psi\}.$$

Traditionally, MI and SI are characterized via increasingly strict levels of invariance that each add further restrictions on statistical parameters across groups. In the classical view, MI is attained only when configural, metric, scalar, and residual invariance hold. These correspond to successive equality restrictions on $\Lambda^{(g)}$, $\nu^{(g)}$, and $\Theta^{(g)}$. More recent perspectives conceptualize invariance as the absence of systematic moderation of key parameters by variables external to the measurement model. Formally, if model parameters depend on group membership through

$$\theta^{(g)} = \theta + \Delta^{(g)},$$

then invariance corresponds to the condition

$$\Delta^{(g)} = 0 \quad \forall g.$$

Covariance invariance analyses evaluate whether the unstandardized covariances among latent factors, represented by elements of $\Psi^{(g)}$, are equivalent across groups, whereas SI analyses examine whether the structural coefficients $\mathbf{B}^{(g)}$ are equal across groups. These analyses are required to determine whether relationships between latent factors are invariant across groups or vary systematically as a function of group membership.

Using Invariance Testing to replicate SEMs

Where MI is conditional on group definitions, replication is conditional on data independence. Formally,

$$\text{Replication} \Longleftrightarrow \theta^{*(d)} \text{ is stable across } d.$$

This definition is formulated at the level of population parameters. Empirically, stability is assessed using the corresponding sample-based parameter estimates $\hat{\theta}^{*(d)}$. This makes the relation to MI and SI explicit: both invariance testing and replication evaluate robustness of parameters under variation, differing mainly in the source of that variation. Replication of SEMs across an original and a replication sample should therefore proceed within a sequential multiple-group manner.

Replication, as defined above, refers to the stability of theory-relevant parameters across independent datasets and is not conditional on measurement invariance. However, in the context of SEM, the empirical assessment of structural parameter stability requires that the underlying measurement model operates equivalently across datasets. Without such equivalence, observed differences in structural parameters may reflect differences in measurement rather than differences in the underlying relations of interest. Measurement invariance is therefore a necessary condition for the valid interpretation of structural replication.

First, measurement invariance between the two samples has to be established:

$$\theta_M^{(1)} = \theta_M^{(2)}.$$

At least metric invariance is required for structural parameters to be meaningfully compared, because without comparable factor loadings, differences in structural relations may reflect measurement non-equivalence rather than substantive divergence. Conditional on sufficient measurement invariance, structural replication is assessed by comparing an unconstrained multiple-group model, in which the structural parameters of interest are freely estimated across datasets, with a constrained model in which they are set to be equal:

$$H_0 : \theta_S^{*(1)} = \theta_S^{*(2)}.$$

This comparison could be evaluated with a likelihood-ratio statistic,

$$\Delta\chi^2 = -2(\ell_C - \ell_U),$$

where ℓ_C and ℓ_U denote the log-likelihoods of the constrained and unconstrained models, respectively. If a model in which the parameters are constrained to be equal across groups fits the data no worse than a model in which they are freely estimated, this is consistent with structural stability of those parameters. Conversely, if the constrained model fits the data worse, this indicates that the parameters differ across groups. At the same time, such tests should be interpreted cautiously because they are sensitive to sample size and do not by themselves capture uncertainty in a substantively calibrated way.

At the level of parameters, again some would consider SEMs replicated when the statistical significance of paths and correlations is consistent across studies. However, failure to reproduce statistical significance does not necessarily imply non-replication, because significance is highly sensitive to sampling variability, statistical power, and measurement error. Replication should therefore not be reduced to consistency of significance decisions alone. This implies that replication of SEMs is best understood as a graded, parameter- and claim-relative process.

Let

$$\theta^* = \{\theta_1^*, \dots, \theta_r^*\}$$

denote the set of parameters relevant to the theoretical claim of interest. Replication can then be represented as the vector

$$\hat{R} = \left(\mathbb{I}\left(\left| \hat{\theta}_j^{*(1)} - \hat{\theta}_j^{*(2)} \right| \leq \epsilon_j \right) \right)_{j=1}^{r},$$

where $\mathbb{I}(\cdot)$ is an indicator function and ϵ_j is a substantively justified tolerance for parameter j. A parameter is said to be empirically *stable* if

$$\left| \hat{\theta}_j^{*(1)} - \hat{\theta}_j^{*(2)} \right| \leq \epsilon_j.$$

Full replication corresponds to the case in which all relevant parameters satisfy this condition, whereas partial replication corresponds to stability of only a subset of them.

This perspective renders approximate and partial replication theoretically grounded rather than ad hoc. Limited deviations in parameter estimates can then be interpreted as meaningful quantitative differences rather than categorical failures of replication, paralleling contemporary approaches to approximate invariance. It follows that models in their entirety should not be treated as the primary objects of replication. Rather, models serve as entry points for evaluating the stability of the theoretical commitments they encode. This shifts attention away from global model fit toward the robustness of substantively meaningful parameters. Because parameter stability under model misspecification may reflect shared misspecification rather than stability of the underlying relations, global fit is a necessary but not sufficient condition for meaningful interpretation of replication. But the converse does not hold. Measurement quality also becomes central to replication validity, because instability across datasets may reflect violations of

$$\theta_M^{(1)} = \theta_M^{(2)}$$

rather than instability of structural relations. Replication claims that do not explicitly address measurement equivalence therefore remain ambiguous.

Conceptual replication has a formal structure under this perspective as it concerns stability of higher-level theoretical claims under deliberate changes in operationalization. This can be understood as stability of $C(\cdot)$ under transformations of the estimand space, rather than equality of the same parameter across datasets. Replication success therefore becomes explicitly conditional on the estimand, the theoretical scope, and the operational context, requiring precise statements about what exactly has or has not replicated. The distinction between invariance testing and replication may therefore be understood as one of scope rather than kind. Invariance evaluates stability across groups within a dataset, whereas replication evaluates stability across independent datasets or study instantiations, but both address the same underlying question of robustness under variation.

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